SLIC 指数 V3

方法论技术白皮书

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作者：依 · 阿卡拉 · 蓬 · 铁曼布拉纳塔姆 副教授

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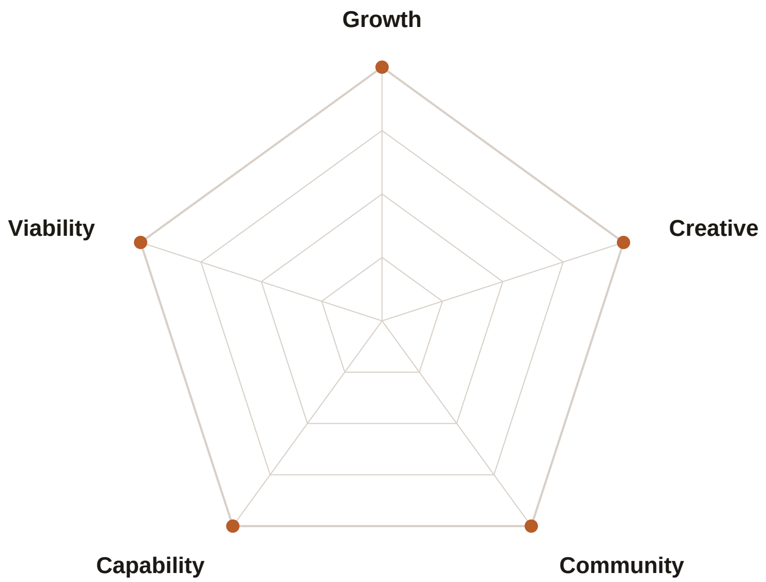
本文档完整记录了 SLIC 指数 V3 的评分方法论：指标定义、归一化锚点表、聚合公式、覆盖度等级，以及 160 座已发布城市的完整排名。



五大支柱。五十点对称。绝对锚点。

执行摘要

SLIC 指数（智慧宜居城市指数）V3 是一个透明、开源的城市排名系统，衡量那些正在城市中建立生活的人们的生活质量——不是游客，不是享受艰苦津贴的外派高管，也不是追逐回报的全球资本。指数覆盖 160 座已发布的城市，数据集以亚太地区为中心但涵盖全球。基于 35 项指标，分为五大支柱进行评分。

五大支柱：

支柱	权重	衡量内容
Growth（增长）	25%	经济活力、可支配收入、创业密度、公民自由
Viability（宜居）	22%	安全、空气质量、医疗、住房可负担性、水资源
Capability（能力）	18%	交通、数字基础设施、教育、可再生能源、可步行性
Community（社区）	15%	包容性、LGBTQ+权利、性别平等、社会信任、收入不平等
Creative（创新）	20%	文化场所、餐饮多样性、艺术资金、创意就业

绝对评分哲学：

- 所有指标均以固定的绝对基准进行归一化——加入第 501 座城市永远不会改变任何现有城市的分数。
- 归一化分数越高始终代表城市表现越好；有害指标在归一化步骤中进行反向评分。
- 调整后的马齐奥塔-帕累托指数（AMPI）聚合公式会惩罚支柱失衡：一座仅有一项出色但其他四项较差的城市，得分低于五项均为中等的城市。
- 缺失数据从不插补。指标数量低于最低要求的城市将获得覆盖度惩罚（－5 或 －15 分）并标注可见等级。
- 每个数据来源都明确标注。每个分数都可追溯至原始数据点、归一化函数与公开数据源。

1.

评分框架

1.1 分段线性归一化

每个原始指标值通过一组固定的绝对锚点映射为 0 – 100 的分数。锚点之间采用线性插值，超出最外侧锚点的值被截断为 0 或 100。对于值越低越好的指标，锚点映射方向反转，使较低的原始值得到更高的归一化分数。

```
s_m(c) = 100 × (winsor(x_m(c), p05_m, p95_m) − p05_m) / (p95_m − p05_m)

Where:
  x_m(c)    = raw metric value for city c on metric m
  p05_m     = 5th percentile across the reference set, frozen at first publication
  p95_m     = 95th percentile, likewise frozen
  winsor()  = clamps to [p05, p95] before normalising
  s_m(c)    = normalized metric score, clamped to [0, 100]

For lower-is-better indicators (homicide, housing burden, hours worked,
etc.), the numerator is inverted so that lower raw values map to higher
normalized scores. Anchors are frozen — adding city #501 never
retroactively changes cities #1–500.
```

1.2 支柱聚合（AMPI）

在每个支柱内，各指标分数使用调整后的马齐奥塔-帕累托指数（AMPI）进行聚合。AMPI 惩罚表现不均：指标间方差较大的城市，得分低于均值相近但方差较小的城市。这使得一个指标的灾难性表现无法通过其他指标的出色表现加以补偿。

```
pillar_score = Σ(s_m × α_(p,m)) / Σ α_(p,m)     [weighted mean]

Where:
  α_(p,m) = published metric weight within pillar p (see Chapter 7)
  s_m     = normalized metric scores for that city, non-null only

Pillar weights are designed so that Σ α_(p,m) within each pillar equals
the pillar's public weight: Pressure 25, Viability 22, Capability 18,
Community 15, Creative 20 (sum = 100).
```

1.3 SLIC 总分

```
Overall SLIC score is aggregated across the five pillar scores using
the Adjusted Mazziotta–Pareto Index (AMPI):

  μ_p   = weighted mean of pillar scores (weights w_p = 25/22/18/15/20)
  σ²_p  = Σ(w_p × (pillar_p − μ_p)²) / Σ w_p          [weighted variance]
  AMPI  = μ_p − σ²_p / μ_p

SLIC(c) = max(0, AMPI(c) − coverage_penalty(c))

AMPI penalises pillar imbalance. A city with four strong pillars and
one catastrophic pillar scores lower than a city with five moderate
pillars — even if their weighted means are identical. This is the
single distinctive claim SLIC makes versus other indices: you cannot
```

```
compensate for a broken pillar with excellence in another. AMPI is
applied at the pillar combination level (n=5, stable variance) rather
than within pillars (where missing metrics would make variance
unreliable).
```

1.4 PPP 调整后的可支配收入（DI_PPP）

SLIC 的标志性指标。衡量扣除所有必要成本后剩余的月度可支配收入，并换算为 PPP 调整后的美元，以便在经济环境差异较大的城市间进行比较。

```
DI_PPP(c) = [
    GrossIncome(c) × (1 − TaxRate(country(c)))
    − Rent(c)
    − Utilities(c)
    − Transit(c)
    − Internet(c)
    − Food(c)
  ] ÷ PPP_PrivateConsumptionFactor(country(c))
```

2.

Pillar 1: Growth

经济活力与机会（权重 25%）

Growth 支柱衡量城市是否为人们建立具有经济生产力的生活创造了条件：扣除成本后的可支配收入、创业与创新密度、公民自由，以及宏观经济动能。

G1 — Real GDP Growth (5-year avg)

Unit: % per annum · Direction: ↑ Higher is better · Source: IMF World Economic Outlook; OECD Regional Statistics

Raw value	Score
≤ 0	0
1.0	25
2.5	50
4.0	75
≥ 6.0	100

G2 — Startup Density

Unit: per 100,000 population · Direction: ↑ Higher is better · Source: Crunchbase; Dealroom

Raw value	Score
≤ 5	0
20	25
50	50
100	75
≥ 200	100

G3 — VC Investment Intensity

Unit: USD per capita, 3-year avg · Direction: ↑ Higher is better · Source: Crunchbase; PitchBook

Raw value	Score
≤ 5	0
50	25
200	50
500	75
≥ 1,500	100

G4 — Ease of Doing Business

Unit: 0–100 index · Direction: ↑ Higher is better · Source: World Bank B-READY Index

Direct passthrough; no transformation required.

Raw value	Score
0	0
25	25
50	50
75	75
100	100

G5 — Civic Freedom

Unit: 0–100 composite · Direction: ↑ Higher is better · Source: Freedom House (0.6) + V-Dem Liberal Democracy Index (0.4)

Composite: 0.6 × FH_rescaled(0–100) + 0.4 × V-Dem_rescaled(0–100). Direct passthrough after composite.

Raw value	Score
0	0
25	25
50	50
75	75
100	100

G6 — Patent Applications

Unit: per 100,000 population, 3-year avg · Direction: ↑ Higher is better · Source: World Intellectual Property Organization (WIPO)

Raw value	Score
≤ 2	0
15	25
40	50
80	75
≥ 150	100

G7 — High-Skill Employment

Unit: % ISCO categories 1–3 · Direction: ↑ Higher is better · Source: International Labour Organization (ILO); LinkedIn Workforce Insights

Raw value	Score
≤ 10	0
20	25
30	50
40	75
≥ 55	100

3.

Pillar 2: Viability

日常宜居与安全（权重 22%）

Viability 支柱衡量城市在日常生活层面是否宜居：人身安全、空气质量、医疗、住房成本负担、水资源获取、气候稳定性与人口活力。

V1 — Homicide Rate

Unit: per 100,000 population · Direction: ↓ Lower is better · Source: UNODC International Homicide Statistics

Lower is better; anchor order is reversed.

Raw value	Score
≥ 50	0
20	25
5	50
1.0	75
≤ 0.3	100

V2 — Air Quality (PM2.5)

Unit: µg/m³ annual mean · Direction: ↓ Lower is better · Source: WHO Global Air Quality Database; IQAir World Air Quality Report

Lower is better; reversed anchors.

Raw value	Score
≥ 80	0
50	25
25	50
10	75
≤ 5	100

V3 — Healthcare Access & Quality

Unit: HAQ Index 0–100 · Direction: ↑ Higher is better · Source: IHME / Lancet Healthcare Access and Quality Index

Direct passthrough.

Raw value	Score
0	0
25	25
50	50
75	75
100	100

V4 — PPP-Adjusted Disposable Income

Unit: USD/month residual · Direction: ↑ Higher is better · Source: Derived: Numbeo cost data + World Bank PPP factors

Non-linear anchors reflect diminishing returns above USD 2,000/month.

Raw value	Score
≤ 0	0
200	15
500	35
1,000	55
2,000	75
≥ 4,000	100

V5 — Housing Burden

Unit: % of gross income · Direction: ↓ Lower is better · Source: Numbeo housing cost surveys; OECD Affordable Housing Database

Lower burden = higher score.

Raw value	Score
≥ 70	0
50	25
30	50
20	75
≤ 10	100

V6 — Water & Sanitation

Unit: % population with safely managed access · Direction: ↑ Higher is better · Source: WHO/UNICEF Joint Monitoring Programme

Raw value	Score
≤ 50	0
70	25
85	50
95	75
≥ 99	100

V7 — Peace & Stability

Unit: Global Peace Index (inverted) · Direction: ↑ Higher is better · Source: Institute for Economics and Peace (IEP)

Formula: score = max(0, min(100, (4 – GPI) / 3 × 100)).

Raw value	Score
GPI ≥ 3.5 → 0	0
GPI 3.0 → 17	~17
GPI 2.0 → 50	50
GPI 1.5 → 83	~83
GPI ≤ 1.2 → 100	100

V8 — Birth Rate

Unit: per 1,000 population · Direction: ↑ Higher is better · Source: UN Population Division; national statistics offices

Interpreted as demographic optimism signal, not population policy judgment.

Raw value	Score
≤ 4	0
7	25
10	50
14	75
≥ 20	100

4.

Pillar 3: Capability

基础设施与系统质量（权重 18%）

Capability 支柱衡量城市基础设施质量：交通覆盖、数字连接、数字治理、教育质量、可步行性、自行车基础设施与可再生能源占比。

C1 — Transit Coverage

Unit: % population within 500m of stop · Direction: ↑ Higher is better · Source: GTFS feeds; ITDP Transit-Oriented Development Database

Raw value	Score
≤ 5	0
20	25
40	50
65	75
≥ 85	100

C2 — Internet Speed

Unit: Mbps median fixed broadband · Direction: ↑ Higher is better · Source: Ookla Speedtest Global Index; M-Lab NDT

Raw value	Score
≤ 5	0
25	25
75	50
150	75
≥ 300	100

C3 — Digital Government

Unit: UN E-Government Development Index × 100 · Direction: ↑ Higher is better · Source: UN Department of Economic and Social Affairs

Direct passthrough.

Raw value	Score
0	0
25	25
50	50
75	75
100	100

C4 — Education Quality

Unit: PISA average score · Direction: ↑ Higher is better · Source: OECD PISA; UNESCO Institute for Statistics (tertiary fallback)

Tertiary enrollment fallback anchors: ≤5%=0, 15%=25, 25%=50, 40%=75, ≥55%=100.

Raw value	Score
≤ 350	0
400	25
450	50
500	75
≥ 550	100

C5 — Walkability

Unit: Walk Score 0–100 · Direction: ↑ Higher is better · Source: Walk Score; OpenStreetMap pedestrian network analysis

Direct passthrough.

Raw value	Score
0	0
25	25
50	50
75	75
100	100

C6 — Cycling Infrastructure

Unit: km of protected lanes per 100,000 pop · Direction: ↑ Higher is better · Source: OpenStreetMap CyclOSM layer; Copenhagenize Index

Raw value	Score
≤ 1	0
5	25
15	50
30	75
≥ 50	100

C7 — Renewable Energy Share

Unit: % of electricity from renewables · Direction: ↑ Higher is better · Source: CDP Cities; IRENA; national grid operators

Raw value	Score
≤ 2	0
15	25
35	50
60	75
≥ 90	100

5.

Pillar 4: Community

归属、包容与社会公平（权重 15%）

Community 支柱衡量城市是否真正开放：LGBTQ+权利、宗教自由、移民融合、收入不平等、社会信任、性别平等，以及周末留驻率（居民是否选择留下的代理指标）。

O1 — LGBTQ+ Rights

Unit: Composite 0–100 · Direction: ↑ Higher is better · Source: ILGA World; Equaldex

Composite: same-sex marriage (25pts) + anti-discrimination law (25pts) + gender recognition (15pts) + no criminalization (20pts) + social acceptance survey (15pts).

Raw value	Score
0	0
25	25
50	50
75	75
100	100

O2 — Religious Freedom

Unit: Pew GRI inverted · Direction: ↑ Higher is better · Source: Pew Research Center Global Restrictions Index

Formula: score = max(0, min(100, (10 – GRI) / 10 × 100)).

Raw value	Score
GRI ≥ 9 → 0	0
GRI 7 → 22	~22
GRI 5 → 44	~44
GRI 2 → 78	~78
GRI ≤ 0.5 → 95	~95

O3 — Immigrant Integration

Unit: Employment gap (percentage points) · Direction: ↓ Lower is better · Source: OECD International Migration Outlook; national labour force surveys

Lower employment gap = higher score.

Raw value	Score
≥ 30 pp	0
20 pp	25
10 pp	50
5 pp	75
≤ 1 pp	100

O4 — Income Inequality (Gini)

Unit: Gini coefficient 0–1 · Direction: ↓ Lower is better · Source: World Bank PovcalNet; OECD Income Distribution Database

Lower Gini = higher score.

Raw value	Score
≥ 0.60	0
0.45	25
0.35	50
0.28	75
≤ 0.22	100

O5 — Social Trust

Unit: % saying most people can be trusted · Direction: ↑ Higher is better · Source: World Values Survey; Gallup World Poll

Raw value	Score
≤ 5%	0
15%	25
30%	50
50%	75
≥ 70%	100

O6 — Gender Equality

Unit: UNDP GII inverted · Direction: ↑ Higher is better · Source: UNDP Human Development Reports — Gender Inequality Index

Formula: score = (1 – GII) × 100.

Raw value	Score
GII 1.0 → 0	0
GII 0.75 → 25	25
GII 0.5 → 50	50
GII 0.25 → 75	75
GII 0.0 → 100	100

O7 — Weekend Retention

Unit: Weekend / weekday mobility ratio · Direction: ↑ Higher is better · Source: Mobile analytics platforms; city-level aggregated movement data

Ratio above 1.0 indicates residents actively choose to stay or return on weekends.

Raw value	Score
≤ 0.70	0
0.80	25
0.90	50
1.00	75
≥ 1.10	100

6.

Pillar 5: Creative

文化丰富度与创意经济（权重 20%）

Creative 支柱衡量居民（而非游客）所经历的文化丰富度：文化场所密度、UNESCO 遗产可达性、餐饮多样性、夜生活、艺术资金、创意就业占比与国际活动密度。

R1 — Cultural Venues

Unit: per 100,000 population · Direction: ↑ Higher is better · Source: OpenStreetMap; Google Places API

Raw value	Score
≤ 2	0
8	25
20	50
40	75
≥ 70	100

R2 — UNESCO World Heritage Sites

Unit: sites within 50km radius · Direction: ↑ Higher is better · Source: UNESCO World Heritage List + GIS distance analysis

Raw value	Score
0	0
1	25
3	50
6	75
≥ 10	100

R3 — Culinary Diversity

Unit: cuisine types per 100,000 population · Direction: ↑ Higher is better · Source: Google Places; Yelp; Foursquare

Raw value	Score
≤ 1	0
3	25
8	50
15	75
≥ 25	100

R4 — Nightlife Density

Unit: bars and clubs per 100,000 population · Direction: ↑ Higher is better · Source: OpenStreetMap; Google Places

Raw value	Score
≤ 3	0
10	25
25	50
50	75
≥ 80	100

R5 — Arts Funding

Unit: USD PPP per capita public arts expenditure · Direction: ↑ Higher is better · Source: Eurostat Culture Statistics; national ministries of culture

Raw value	Score
≤ 5	0
30	25
80	50
150	75
≥ 300	100

R6 — Creative Employment

Unit: % of workforce in creative industries · Direction: ↑ Higher is better · Source: UNCTAD Creative Economy Report; national labour force surveys

Raw value	Score
≤ 1%	0
3%	25
6%	50
10%	75
≥ 15%	100

R7 — International Events

Unit: per million population per year · Direction: ↑ Higher is better · Source: ICCA International Congress and Convention Association; Eventbrite

Raw value	Score
≤ 2	0
10	25
25	50
50	75
≥ 100	100

7.
数据来源

SLIC 采用五层数据源层级。第 1 层（城市官方）优先使用；回退顺序为：第 2 层（次国家级）→ 第 3 层（国家/国际级）→ 第 4 层（经审核的次级/实验来源）→ 第 5 层（分析师评估）。每个已发布指标的数据源层级显示在城市评分卡页面上。

层级	类别	示例
1	城市 / 都会区官方	Municipal open data portals, utility feeds, transit GTFS
2	次国家级官方	State, provincial, or regional government data
3	国家 / 国际级官方	World Bank, WHO, ILO, UNESCO, OECD, WIPO, UN Agencies
4	经审核的次级与实验来源	OpenAQ, Copernicus CAMS, M-Lab NDT, satellite remote sensing
5	分析师评估	SLIC analyst cross-reference and lived-experience review

主要数据源机构

- World Bank — WDI, B-READY, PovcalNet, PPP conversion factors
- World Health Organization — GHO OData API, HAQ Index, JMP water/sanitation
- International Labour Organization — ILOSTAT (employment, hours, wages)
- OECD — PISA, Health Statistics, Affordable Housing, Income Distribution
- UN Population Division — birth rates, demographic projections
- UNESCO Institute for Statistics — education enrollment, cultural indicators
- UNODC — International Homicide Statistics
- UNDP — Human Development Reports, Gender Inequality Index
- IEP — Global Peace Index
- Freedom House — Freedom in the World annual scores
- V-Dem Institute — Liberal Democracy Index
- Pew Research Center — Global Restrictions on Religion Index
- WIPO — Patent Application Statistics
- IMF — World Economic Outlook GDP data
- IRENA — Renewable capacity and generation statistics
- IQAir / WHO — PM2.5 ambient air quality annual averages
- ILGA World + Equaldex — LGBTQ+ legal environment composite
- World Values Survey + Gallup — social trust indicators
- ICCA — International Congress and Convention Association statistics
- Numbeo — Cost of living, housing, and consumer price surveys
- Crunchbase / PitchBook / Dealroom — startup and VC investment data
- Ookla + M-Lab — broadband speed measurements
- OpenStreetMap / CyclOSM — cycling and pedestrian infrastructure
- Walk Score — walkability index

- OpenAQ / Copernicus CAMS — supplementary air quality monitoring

8.

覆盖度等级

SLIC 不对缺失值进行插补。当指标不可用时，该指标将从支柱聚合中排除，并应用覆盖度惩罚。覆盖度等级与每个城市的分数一同发布。

等级	条件	惩罚	解读
A	7 项指标中 ≥ 6 项（或 5 – 6 项指标支柱中 ≥ 5 项）	无	对支柱得分完全可信
B	≥ 4 项指标	− 5 分	得分可靠，但需审慎解读
C	≤ 3 项指标	− 15 分，标为临时值	仅作参考；存在显著数据缺口

惩罚在支柱级 AMPI 计算后施加，并以 0 为下限。整体覆盖度等级反映该城市最差的支柱等级。城市不会因为覆盖度低而被隐藏——低覆盖度会被显现，而非掩盖。

9.

已发布城市排名

所有 138 座已发布城市按 SLIC 总分排序。分数四舍五入到一位小数。允许并列——四舍五入后分数相同的城市共享同一排名。支柱列标题：G = Growth, V = Viability, Cap = Capability, Com = Community, Cr = Creative。

#	城市	国家	SLIC	G	V	Cap	Com	Cr	等级
1	Raleigh	United States	66.6	70.1	84.1	79.1	43.5	62.0	A
2	Kaohsiung	Taiwan	66.1	63.2	86.5	90.6	48.1	56.9	A
3	Montreal	Canada	65.7	57.3	93.3	75.3	64.4	54.1	A
3	Singapore	Singapore	65.7	50.4	87.4	94.1	47.8	73.3	A
5	Minneapolis	United States	65.6	66.1	84.1	79.1	43.5	62.0	A
6	Pittsburgh	United States	65.3	71.5	84.1	79.1	43.5	55.7	A
6	Abu Dhabi	United Arab Emirates	65.3	65.4	92.6	62.0	78.8	46.2	A
8	Dubai	United Arab Emirates	65.0	64.4	92.6	62.0	78.8	46.2	A
9	Adelaide	Australia	64.1	60.3	89.5	87.1	53.2	49.2	A
10	Melbourne	Australia	63.3	49.1	89.5	87.1	53.2	60.5	A
11	Manama	Bahrain	63.0	82.6	79.7	57.8	62.9	42.5	A
12	Ottawa	Canada	62.7	49.3	93.3	75.3	64.4	54.1	A
12	Hobart	Australia	62.7	60.3	89.5	87.1	53.2	45.1	A
14	Graz	Austria	62.1	57.9	87.1	74.4	70.7	41.4	A
14	Perth	Australia	62.1	52.9	89.5	87.1	53.2	51.2	A
16	Taipei	Taiwan	62.0	43.7	89.5	89.7	48.1	69.4	A
17	Eindhoven	Netherlands	61.8	59.6	97.6	71.8	60.9	42.8	A
18	Brisbane	Australia	61.6	49.1	89.5	87.1	53.2	54.4	A
19	Haifa	Israel	60.6	64.1	86.1	64.7	41.6	54.7	A
20	Chicago	United States	59.8	48.4	84.1	79.1	43.5	62.0	A
21	Dunedin	New Zealand	59.0	52.6	92.6	85.7	58.4	39.1	A
22	Milan	Italy	58.8	88.5	85.8	66.9	50.8	29.4	A
23	Lyon	France	58.5	47.5	97.9	63.9	72.0	45.1	A
24	Valparaiso	Chile	58.3	56.3	72.7	83.8	49.8	44.6	A
25	Christchurch	New Zealand	58.2	46.3	92.6	85.7	58.4	43.6	A
26	Khobar	Saudi Arabia	57.4	88.5	96.3	78.9	42.8	24.1	A
27	Jeddah	Saudi Arabia	57.2	87.3	96.3	78.9	42.8	24.1	A
28	Braga	Portugal	57.0	54.5	95.7	68.8	47.3	43.8	A
28	Riyadh	Saudi Arabia	57.0	86.1	96.3	78.9	42.8	24.1	A
30	Jeju City	South Korea	56.3	53.8	99.2	85.8	30.0	50.6	A
31	New York	United States	56.2	23.9	85.4	82.0	71.5	75.3	A
32	London	United Kingdom	56.1	30.1	82.8	78.4	74.2	59.3	A
33	Copenhagen	Denmark	55.9	26.0	97.9	71.3	89.3	61.5	A
34	Sydney	Australia	55.6	31.0	89.5	87.1	53.2	63.9	A
34	Wellington	New Zealand	55.6	36.8	92.6	85.7	58.4	48.8	A
36	Torun	Poland	55.5	66.7	79.5	65.1	51.2	31.9	A
37	Vienna	Austria	55.3	40.2	87.1	74.4	70.7	41.4	A

#	城市	国家	SLIC	G	V	Cap	Com	Cr	等级
38	Santiago	Chile	55.2	47.2	72.7	83.8	49.8	44.6	A
38	Zurich	Switzerland	55.2	33.7	97.6	72.7	72.8	49.4	A
40	Busan	South Korea	55.1	50.8	99.2	85.8	30.0	50.6	A
41	Suwon	South Korea	55.0	50.5	93.1	85.8	30.3	50.6	A
41	Incheon	South Korea	55.0	50.5	99.2	85.8	30.0	50.6	A
41	Toronto	Canada	55.0	32.4	93.3	75.3	64.4	54.1	A
44	Helsinki	Finland	54.9	33.8	91.5	81.0	58.2	52.9	A
45	Katowice	Poland	54.8	63.9	79.5	65.1	51.2	31.9	A
46	Porto	Portugal	54.5	47.5	95.7	68.8	47.3	43.8	A
47	Tallinn	Estonia	53.6	45.5	90.1	58.4	38.6	57.8	A
47	Auckland	New Zealand	53.6	30.6	92.6	85.7	58.4	52.9	A
49	Gdansk	Poland	53.4	58.2	79.5	65.1	51.2	31.9	A
50	Vancouver	Canada	53.3	29.3	93.3	75.3	64.4	54.1	A
50	Tokyo	Japan	53.3	39.7	94.2	76.8	51.6	44.2	A
52	Tel Aviv	Israel	53.1	41.4	86.1	64.7	41.6	54.7	A
53	Fukuoka	Japan	50.9	57.0	94.5	67.4	25.8	42.3	A
53	Bratislava	Slovakia	50.9	54.1	89.7	49.7	51.5	35.3	A
53	Krakow	Poland	50.9	49.6	79.5	65.1	51.2	31.9	A
56	Bologna	Italy	50.8	50.9	88.2	66.9	50.8	29.4	A
57	Amsterdam	Netherlands	49.9	31.4	97.6	71.8	60.9	42.8	A
58	Hiroshima	Japan	49.1	57.0	94.5	67.4	25.8	37.3	A
58	Sapporo	Japan	49.1	57.0	94.5	67.4	25.8	37.3	A
60	Doha	Qatar	49.0	53.9	73.9	54.4	36.5	37.6	A
61	Kuwait City	Kuwait	48.8	87.2	69.2	65.0	40.5	19.9	A
62	Kobe	Japan	48.7	55.6	94.5	67.4	25.8	37.3	A
63	Bangkok	Thailand	47.4	43.4	76.6	56.4	55.9	30.0	A
64	Port Louis	Mauritius	47.2	54.6	76.2	47.0	26.2	45.4	A
64	Melaka	Malaysia	47.2	63.0	75.3	52.4	31.4	31.5	A
66	Kuching	Malaysia	47.0	61.3	73.7	52.4	32.2	31.5	A
67	Kota Kinabalu	Malaysia	46.9	61.3	77.9	52.4	31.1	31.5	A
68	Salalah	Oman	46.5	72.0	71.1	42.0	35.9	30.6	A
69	Paris	France	46.3	23.1	97.9	63.9	72.0	45.1	A
69	Montevideo	Uruguay	46.3	49.6	59.5	74.6	50.5	24.4	A
71	George Town	Malaysia	46.1	57.2	73.7	52.4	32.2	31.5	A
72	Tianjin	China	45.6	40.6	98.6	71.9	32.4	37.1	A
73	Muscat	Oman	45.5	63.6	71.1	42.0	35.9	30.6	A
74	Bucharest	Romania	44.9	53.6	81.4	50.6	27.6	34.8	A
75	Budapest	Hungary	44.7	42.9	93.1	55.5	52.3	26.2	A
76	San Juan	Puerto Rico	44.6	35.3	59.2	68.3	61.3	45.8	B
76	Cordoba	Argentina	44.6	65.5	69.2	78.6	49.0	10.2	A
78	San Jose	Costa Rica	44.4	40.2	43.5	66.0	54.6	35.9	A
79	Chongqing	China	44.2	36.5	75.2	71.9	33.0	37.1	A
80	Kuala Lumpur	Malaysia	44.1	48.9	73.7	52.4	32.2	31.5	A
81	Belgrade	Serbia	44.0	43.4	74.8	53.2	39.7	30.3	A

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82	Chengdu	China	43.8	35.4	77.0	71.9	33.2	37.1	A
83	Hat Yai	Thailand	43.5	55.1	66.7	56.4	36.4	22.8	A
84	Male	Maldives	42.4	54.1	72.1	39.8	26.5	35.4	A
85	Guangzhou	China	41.9	33.0	98.6	71.9	32.4	37.1	A
86	Phuket	Thailand	41.5	46.6	68.0	56.4	36.4	22.8	A
87	Panama City	Panama	41.4	55.5	44.0	64.2	33.0	26.9	A
88	Chiang Mai	Thailand	41.1	54.3	49.7	56.4	36.4	22.8	A
89	Hangzhou	China	40.9	29.2	84.0	71.9	33.3	37.1	A
89	Buenos Aires	Argentina	40.9	50.7	69.2	78.6	49.0	10.2	A
91	Shenzhen	China	39.9	29.2	98.6	71.9	32.4	37.1	A
92	Curitiba	Brazil	38.2	49.4	45.1	69.5	23.2	27.6	A
93	Florianopolis	Brazil	37.8	47.6	45.1	69.5	23.2	27.6	A
94	Arequipa	Peru	37.7	55.9	34.1	42.3	44.0	24.7	A
95	Nizhny Novgorod	Russia	37.6	70.2	62.4	59.2	21.1	15.5	A
96	Shanghai	China	37.2	24.4	98.6	71.9	32.4	37.1	A
97	Lima	Peru	36.8	47.8	34.1	42.3	44.0	24.7	A
98	Sao Paulo	Brazil	35.0	37.6	45.1	69.5	23.2	27.6	A
99	Medellin	Colombia	34.4	52.4	28.0	66.3	40.0	19.6	A
100	Kigali	Rwanda	33.7	54.3	47.2	15.8	53.3	24.8	A
101	Rabat	Morocco	33.4	45.5	59.9	34.1	28.9	18.8	A
101	Windhoek	Namibia	33.4	41.6	30.5	32.1	46.9	25.4	A
103	Casablanca	Morocco	33.2	44.3	59.9	34.1	28.9	18.8	A
103	Cebu City	Philippines	33.2	55.0	38.2	40.4	30.9	17.5	A
105	Bogota	Colombia	33.0	45.1	28.0	66.3	40.0	19.6	A
106	Makati	Philippines	32.8	50.1	38.2	40.4	30.9	17.5	A
107	Surabaya	Indonesia	32.1	63.2	40.8	35.1	27.6	17.8	A
107	Jakarta	Indonesia	32.1	56.6	40.8	35.1	27.6	17.8	A
107	Gaborone	Botswana	32.1	50.3	31.9	22.2	25.1	39.0	A
110	Moscow	Russia	30.4	35.2	62.4	59.2	21.1	15.5	A
111	Suva	Fiji	30.2	41.6	36.1	47.7	13.6	25.1	A
112	Aqaba	Jordan	28.2	41.5	66.0	46.5	10.8	17.0	A
113	Hyderabad	India	28.1	43.9	53.1	30.4	20.8	14.6	A
114	Pune	India	28.0	43.6	53.1	30.4	20.8	14.6	A
115	Bengaluru	India	27.9	42.7	53.1	30.4	20.8	14.6	A
116	Chandigarh	India	27.8	43.4	44.1	30.4	21.5	14.6	A
117	Amman	Jordan	27.0	37.5	66.0	46.5	10.8	17.0	A
118	Merida	Mexico	26.1	49.8	13.3	55.5	55.5	14.1	A
119	Kandy	Sri Lanka	25.4	51.5	40.7	33.7	15.8	11.7	A
119	Cape Town	South Africa	25.4	31.5	15.1	26.3	53.6	29.8	A
119	Colombo	Sri Lanka	25.4	48.3	40.7	33.7	15.8	11.7	A
119	Johannesburg	South Africa	25.4	31.7	15.1	26.3	53.6	29.8	A
123	Guadalajara	Mexico	25.3	45.0	13.3	55.5	55.5	14.1	A
124	Nairobi	Kenya	24.0	45.6	45.1	12.5	29.3	14.5	A
124	Thimphu	Bhutan	24.0	46.7	47.0	25.2	18.9	10.2	A

#	城市	国家	SLIC	G	V	Cap	Com	Cr	等级
126	Accra	Ghana	23.4	53.3	35.9	14.9	21.7	17.8	A
127	Kumasi	Ghana	23.0	58.7	35.9	14.9	21.7	17.8	A
128	Chattogram	Bangladesh	22.7	51.3	48.3	10.8	23.3	15.7	A
128	Dhaka	Bangladesh	22.7	48.8	48.3	10.8	23.3	15.7	A
130	Pokhara	Nepal	22.1	44.3	37.6	12.2	20.9	16.6	A
130	Kathmandu	Nepal	22.1	41.7	37.6	12.2	20.9	16.6	A
132	Mexico City	Mexico	22.0	32.1	13.3	55.5	55.5	14.1	A
133	Da Nang	Vietnam	20.2	51.0	—	44.3	—	20.8	C
134	Phnom Penh	Cambodia	15.4	41.9	—	18.2	—	36.3	C
135	Dakar	Senegal	12.4	38.8	0.7	10.6	45.1	26.7	A
136	Islamabad	Pakistan	11.6	47.2	32.7	8.1	15.3	4.4	A
137	Karachi	Pakistan	11.5	48.0	32.7	8.1	15.3	4.4	A
138	Lahore	Pakistan	11.4	48.3	32.7	8.1	15.3	4.4	A

10.

设计原则

绝对评分

锚点固定于现实世界的阈值。城市分数不会因新城市加入指数而改变。加入第 501 座城市不会改变第 1 – 500 座城市的分数。

透明可追溯

每个已发布分数都可追溯至原始数据点、归一化函数与具名来源。评分工作簿可公开下载。

惩罚失衡

AMPI 聚合公式意味着一座城市无法通过在其他支柱上的出色表现来弥补某一支柱上的灾难性表现。极端失衡将在结构上受到惩罚。

坦诚面对数据缺口

缺失数据从不插补。缺口通过可见的覆盖度等级与惩罚加以标示。低覆盖度城市被显示，而非隐藏。

可扩展且无需修订

可以将新城市加入指数而无需重新计算现有分数。当城市宇宙扩展时，方法论无需回溯修订。

抗套路化

指数的设计使城市难以通过仅优化可见代理指标来操纵分数。DI_PPP 公式、AMPI
惩罚与失衡惩罚都难以在不真正改善城市状况的情况下被操控。

11.

符号术语表

符号 / 术语	定义
c	正在评估的城市或功能城市区域
m	指标索引（支柱内的单个指标）
p	支柱索引（五大主题支柱之一）
x_m(c)	城市 c 在指标 m 上的原始观测值
s_m(c)	城市 c 在指标 m 上归一化后的分数，范围 [0, 100]
μ	支柱内或跨支柱归一化分数的算术平均值
σ	支柱内或跨支柱分数的总体标准差
cv	变异系数：σ / μ
AMPI	调整后的马齐奥塔-帕累托指数：μ − σ ²/μ
p05_m, p95_m	指标 m 的 5 与 95 分位边界，于首次发布时冻结
DI_PPP(c)	城市 c 扣除必要成本后经 PPP 调整的月度可支配收入
α_(p,m)	支柱 p 内指标 m 的权重
γ_m	指标 m 的覆盖度权重（用于覆盖率计算）
w_p	公开支柱权重（Growth 0.25, Viability 0.22, Capability 0.18, Community 0.15, Creative 0.20）
HAQ	医疗可及性与质量指数（IHME/Lancet）
GPI	全球和平指数（经济与和平研究所）
GII	性别不平等指数（UNDP）
GRI	政府限制指数（皮尤研究中心）
EGDI	电子政务发展指数（联合国）
PPP	购买力平价——在不同货币间等化购买力的转换因子

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